How Virtualization Enabled the Growth of Cloud Service Providers

CSA1522-Cloud Computing and Big Data Analytics

CO1_AT2- Technical Presentation

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What Is Virtualization?

- Creating a software-based version of a server, storage device, or network
- Lets multiple isolated environments (VMs) run on one physical machine.
- Key ideas: abstraction, isolation, and efficient resource sharing.
- Virtualization creates software-based versions of servers, storage, and network resources instead of using only physical hardware.
- Virtualization reduces infrastructure costs by minimizing the need for additional physical servers.

The Hypervisor

- A hypervisor is software that creates, runs, and manages virtual machines on a physical host.
- It allocates CPU, memory, storage, and network resources to each virtual machine.
- It ensures that virtual machines remain isolated for security and stable performance.
- The hypervisor monitors resource usage and balances workloads efficiently.
- It allows multiple operating systems to run simultaneously on the same hardware.
- Hypervisors are classified into Type 1 (Bare Metal) and Type 2 (Hosted).

Type 1 — Bare Metal Hypervisor

- A Type 1 hypervisor is installed directly on the physical hardware without a host operating system.
- It provides near-native performance because there is no additional software layer.
- It offers strong security and isolation between virtual machines.
- It is widely used in enterprise data centers and public cloud environments.
- It efficiently manages hardware resources for large-scale virtualization.
- Popular examples include VMware ESXi, Microsoft Hyper-V, Xen, and KVM.

Type 2 — Hosted Hypervisor

- A Type 2 hypervisor runs as an application on top of an existing operating system.
- It is simple to install and ideal for personal computers and laptops.
- It is commonly used for software testing, development, and educational purposes.
- Performance is slightly lower because it depends on the host operating system.
- It supports running multiple operating systems without changing the physical computer.
- Examples include Oracle VirtualBox, VMware Workstation, and Parallels Desktop.

Bare Metal vs. Hosted Hypervisor

Aspect	Type 1 — Bare Metal	Type 2 — Hosted
Installation	Directly on physical hardware	On top of an existing host OS
Performance	Near-native, low overhead	Reduced due to host OS layer
Typical use	Data centers, cloud platforms	Desktops, development/testing
Security	Smaller attack surface	Inherits host OS vulnerabilities
Examples	VMware ESXi, Hyper-V, Xen, KVM	VirtualBox, Workstation, Parallels

How Virtualization Enabled Cloud Growth

- Virtualization enables multiple customers to securely share the same physical server through multi-tenancy.
- It allows virtual machines to be created, modified, or removed within minutes.
- Resource pooling helps providers efficiently share computing, storage, and networking resources.
- Higher hardware utilization significantly reduces infrastructure and maintenance costs.
- Virtualization makes cloud services scalable by supporting rapid expansion of resources.
- It provides the flexibility required for on-demand cloud computing services.

Key Benefits for Cloud Providers

- Virtualization increases server utilization, reducing hardware waste and operational expenses.
- It enables rapid deployment of virtual machines without purchasing new hardware.
- Providers can offer pay-as-you-go pricing by measuring virtual resource usage.
- Virtual machines can be migrated or restored quickly for disaster recovery purposes.
- Standardized VM templates simplify deployment across different data centers.
- Cloud providers can easily scale services by adding more physical hosts to the virtualization platform.

Real-World Examples

- Amazon Web Services (AWS) uses the Nitro Hypervisor to deliver secure and high-performance virtual machines.
- Microsoft Azure relies on the Hyper-V hypervisor to manage its cloud infrastructure.
- Google Cloud Platform uses a customized KVM hypervisor across its global data centers.
- These providers use virtualization to support millions of users simultaneously.
- Virtualization improves resource efficiency while maintaining security and reliability.
- It enables cloud providers to deliver scalable and cost-effective computing services worldwide.

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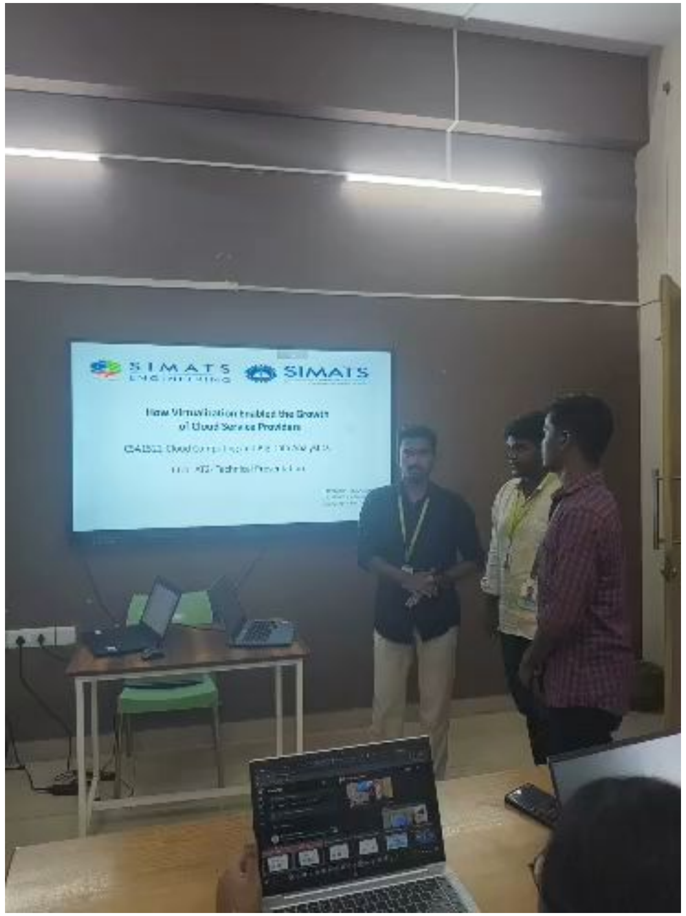
Conclusion

- Virtualization has transformed traditional data centers into flexible cloud environments.
- It improves resource utilization while reducing hardware and maintenance costs.
- The technology provides scalability, flexibility, and efficient resource management.
- Hypervisors make it possible to run multiple virtual machines securely on one server.
- Virtualization is the foundation of all major cloud service providers today.
- Future technologies such as Docker containers, Kubernetes, and serverless computing continue to build on virtualization.

<u>Geo Tag Picture</u>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